

Problem

The transfer function of a linear system is

$$H(s) = \frac{2s}{s+6}$$

Find the output $y(t)$ due to the input $45e^{-3t}u(t)$ and its impulse response.

Step-by-step solution

Step 1 of 4

The Laplace transform of the input is,

$$X(s) = \frac{45}{s+3}$$

The expression for the transfer function of the system is,

$$H(s) = \frac{Y(s)}{X(s)}$$

The output response is,

$$Y(s) = H(s)X(s)$$

Substitute $\frac{2s}{s+6}$ for $H(s)$ and $\frac{45}{s+3}$ for $X(s)$.

$$\begin{aligned} Y(s) &= \left(\frac{2s}{s+6} \right) \left(\frac{45}{s+3} \right) \\ &= \frac{90s}{(s+6)(s+3)} \end{aligned}$$

Step 2 of 4

Apply partial fractions.

$$Y(s) = \frac{A}{s+6} + \frac{B}{s+3} \dots\dots (1)$$

$$\frac{90s}{(s+6)(s+3)} = \frac{A}{s+6} + \frac{B}{s+3} \dots\dots (2)$$

Put $s = -6$ in equation (2).

$$\begin{aligned} A &= \frac{90(-6)}{(-6+3)} \\ &= 180 \end{aligned}$$

Put $s = -3$ in equation (2).

$$\begin{aligned} B &= \frac{90(-3)}{(-3+6)} \\ &= -90 \end{aligned}$$

Step 3 of 4

Substitute 180 for A and -90 for B in equation (1).

$$Y(s) = \frac{180}{s+6} - \frac{90}{s+3}$$

Apply inverse Laplace transform.

$$\begin{aligned} y(t) &= 180e^{-6t}u(t) - 90e^{-3t}u(t) \\ &= (180e^{-6t} - 90e^{-3t})u(t) \\ &= (-90e^{-3t} + 180e^{-6t}), \quad t \geq 0 \end{aligned}$$

Thus, the output $y(t)$ is $\boxed{(-90e^{-3t} + 180e^{-6t}), \quad t \geq 0}$.

Step 4 of 4

The Laplace transform of the impulse function is 1.

$$X(s) = 1$$

Find the impulse response of the system.

$$Y(s) = H(s)X(s)$$

Substitute $\frac{2s}{s+6}$ for $H(s)$ and 1 for $X(s)$.

$$\begin{aligned} Y(s) &= \frac{2s}{s+6} \\ &= 2 \left(1 - \frac{6}{s+6} \right) \end{aligned}$$

Apply inverse Laplace transform.

$$\begin{aligned} y(t) &= 2[\delta(t) - 6e^{-6t}u(t)] \\ &= 2\delta(t) - 12e^{-6t}u(t) \end{aligned}$$

Thus, the impulse response of the system is $\boxed{2\delta(t) - 12e^{-6t}u(t)}$.